

Nigar Najafli

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EDUCATION

University of Maryland-College Park

College Park, MD/United States

PhD in Electrical and Computer Engineering

August 2026 - May 2031

- Current course work: Digital Computer Design, Mathematical Foundations for Computer Engineering

Carnegie Mellon University

Pittsburgh, PA/United States

Master of Science in Information Security

August 2024 - May 2026

- Relevant course work: Applied Information Assurance, Introduction to Computer Systems, Hacking and Offensive Security, Security in Networked Systems, Browser Security, Advanced Real-World Data Networks, DevSecOps, Cyber Risk Modeling
- GPA: 4.0/4.0

Baku Higher Oil School

Baku, Azerbaijan

Bachelor's Degree in Information Security

September 2019 - July 2024

- Relevant course work: Web Programming and Security, Mobile Programming and Security, Ethical Hacking and Defense, Quality Assurance Management, Network Security, Information Security Strategy and Policy, Artificial Intelligence and Machine Learning Fundamentals, Attacks on Cryptosystems, Database Administration and Security, Forensic Science
- Taught Red Team classes for younger students in university's Cyber Club.
- GPA: 96.06/100.

RESEARCH EXPERIENCE

Research Contributor - [PVC: Private Verifiable Credentials with Minimal Adoption Barriers](#)

Carnegie Mellon University – Collaboration with Prof. Scheffler

July 2025 – November 2025

- Contributed to the development of a Circom-based zero knowledge circuit enabling private proof of age, signature verification, and challenge freshness without revealing underlying personal data.
- Gained practical experience in applied cryptography, zk-SNARK design, and privacy-preserving authentication workflows.

Undergraduate Thesis Research - Active Directory Security and Hardening

Baku Higher Oil School

April 2024 – May 2024

- Conducted a full security assessment of an enterprise-grade Active Directory environment.
- Built a controlled AD lab on Hyper-V using AutomatedLab and populated it with realistic misconfigurations.
- Produced a structured hardening framework combining backup strategy, privileged access workstations, enhanced auditing, and protocol hardening.

Independent Research - [Building an Effective Security Operation Center with Open-Source Tools](#)

Baku Higher Oil School

April 2023 – May 2023

- Designed and implemented a fully functional Security Operations Center (SOC) using open-source technologies.
- Performed simulated attack exercises to assess SOC responsiveness and proposed improvements to analyst workflow, detection coverage, and system tuning.

PROJECTS

[LLM Jailbreak & Prompt Injection Red-Team Harness](#)

May 2026

- Built an LLM red-team harness that benchmarks prompt-injection defenses across three open models, using canary-token and LLM-judge scoring mapped to OWASP LLM Top 10 and ATLAS to show no defense holds against adaptive attacks.

[ThreatFlow – Real-Time & Batch Network Threat Analytics Platform](#)

March 2026

- Deployed a real-time + batch threat analytics pipeline on GCP (GKE, Kafka, Spark Streaming, Dataproc, Terraform), streaming ~100 events/sec to a live Streamlit dashboard with Prometheus/Grafana observability and clean SonarQube ratings.

[CVE-2022-45782 - dotCMS Password-Reset Token Predictability \(PoC\)](#)

November 2025

- Developed a proof-of-concept for CVE-2022-45782, demonstrating full admin account takeover in dotCMS by exploiting predictable, non-cryptographically generated password-reset tokens and brute-forcing the token seed with a multithreaded cracker.

Evaluating browser fingerprinting defenses

April 2025

- Evaluated major anti-tracking tools using CoverYourTracks to assess resistance to high-entropy fingerprint attributes.
- Ran structured fingerprinting experiments in a controlled Ubuntu VM to compare privacy tool performance and identify weaknesses in normalization-based defenses.

Evaluating the Security of DNS Protocols with a CORE-based Framework

April 2025

- Built a full DNS testing setup in CORE emulator, including web servers, DNS resolver, root DNS server, and attacker nodes.
- Configured DNS infrastructure using Bind9, Docker-isolated servers, custom zone files, TLS certificates, and DoT support.
- Developed three attack scripts to test the Confidentiality, Integrity, and Availability guarantees of standard DNS and DoT.

Facial Expression Recognition Pipeline

May 2023

- Built a facial expression classifier using transfer learning (VGG16/AlexNet) with custom splits, transforms and loaders.
- Trained and fine-tuned the model with a modular PyTorch pipeline with configurable hyperparameters and checkpointing.

CyberLearn: CTF & Cybersecurity Learning Platform for High School Students in Azerbaijani language

September 2021

- Designed and developed CyberLearn, an interactive cybersecurity and CTF platform for beginners and high school students.
- Built challenge modules covering basic networking, web security, and cryptography concepts.

CERTIFICATIONS

[OSWE](#) & [OSCP](#) by OffSec, Certified DevSecOps Professional ([CDP](#)) by Practical DevSecOps, Certified Red Team Operator ([CRTO](#)) by Zero-Point Security, Certified Ethical Hacker (CEH) Practical by EC-Council

WORK EXPERIENCE

PASHA Bank OJSC

Baku, Azerbaijan

Application Security and Penetration Testing Specialist

April 2023 - August 2024

- Conducted penetration testing on over 15 web applications and 4 mobile applications of bank.
- Collaborated with 10+ development teams to remediate over 200 identified vulnerabilities through secure code reviews.
- Configured SAST tool in the CI/CD pipeline automating static security scans for ~50k lines of code.
- Evaluated results from SCA, SAST, and DAST tool scans of over 700 repositories to identify and address vulnerabilities.
- Tuned and optimized SAST tool rules to reduce false positive rate to less than 5%, improving development team accuracy.

PROSOL CJSC

Baku, Azerbaijan

Junior Cyber Defense Specialist

June 2022 - March 2023

- Performed administration, testing, and integration tasks related to SIEM platforms for 5 customers.
- Created and developed correlation and detection rules to reduce false positives by 30%.
- Automated incident response procedures through SOAR to improve incident response time by 20%.
- Developed automated tool to analyze Indicators of Compromise (IOCs) across multiple Threat Intelligence Platforms upon incident generation and send detailed reports to analysts for investigation.

AWARDS

Distinguished Participation at the Digital Transformation - 2023 Conference

Baku, Azerbaijan

Baku Higher Oil School

June 2023

- Research topic: [Building an Effective Security Operation Center with Open-Source Tools](#)

Presidential Scholarship Recipient

Baku, Azerbaijan

Government of Azerbaijan

September 2019

- Awarded full Presidential Scholarship for scoring 680.3/700 on the national university entrance examination in Azerbaijan (ranking among the top students nationwide).

ACTIVITIES

Carnegie Mellon University - INI

Pittsburgh, PA/United States

Teaching Assistant for Introduction to Information Security

August 2025 - Present

- Assist professor with classroom instructions, exams, assignments and mentored students.

DEF CON 33 / Adversary Village

Las Vegas, NV/United States

Volunteer

August 2025

- Assisted with workshops, coordinated attendees, and supported security researchers during hands-on sessions.

SKILLS

Cybersecurity: Web/Mobile/Network Penetration Testing, DevSecOps, Security Engineering, Secure Code Review, Threat Modeling

Penetration Testing: Burp Suite Professional, Qualys, Invicti, Tenable, Frida, MobSF, PowerSploit, Active Directory, Metasploit

Security Operations (SOC): Cortex XDR, TheHive, Shuffle, Splunk, Security Onion, Wazuh, ELK Stack, DLP, SIEM

CI/CD & DevSecOps: Snyk, Semgrep, Coverity, Prisma Cloud, Docker, DefectDojo, OWASP ZAP, GitHub Actions, IriusRisk

Programming & Scripting: Python, JavaScript, SQL, Bash, C/C++, PowerShell, Java.